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Machine Learning (410242)

BE (Computer Engineering) — Semester VII — 2019 Pattern

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Figures to the right indicate full marks.
3. Neat diagrams must be drawn wherever necessary.
4. Make suitable assumptions whenever necessary.

Unit III — Supervised Learning: Regression — 18 Marks

Q1) Solve **any one** of the following:

a) Explain Linear Regression in detail. Derive the closed-form solution for the Ordinary Least Squares (OLS) estimate. [6]

b) Differentiate between overfitting and underfitting with suitable examples. How does regularization help mitigate overfitting? [6]

c) Write a short note on evaluation metrics for regression: MAE, RMSE, and R^2. Compare their properties. [6]

OR

Q2) Solve **any one** of the following:

a) Explain Lasso (L1) and Ridge (L2) regression. How does the regularization parameter λ affect the model coefficients? [6]

b) The following table shows the number of study hours (X) and exam scores (Y) for 5 students: [6]

X (Hours)	2	4	6	8	10
Y (Score)	55	60	68	75	82

Find the regression line $Y = a + bX$ using the least squares method. Estimate the score for 12 hours of study.

c) What is the Gradient Descent algorithm? Explain batch, stochastic, and mini-batch gradient descent. [6]

Unit IV — Supervised Learning: Classification — 17 Marks

Q3) Solve **any one** of the following:

a) Explain the K-Nearest Neighbors (KNN) algorithm with a suitable example. How does the choice of K and the distance metric affect classification performance? [5]

b) What is Ensemble Learning? Explain Bagging and Boosting with examples. How does Random Forest
combine these concepts? [6]

c) Write a short note on metrics for evaluating classifier performance: accuracy, precision, recall, F1-score, and confusion matrix. [6]

OR

Q4) Solve **any one** of the following:

a) Compare Binary Classification with Multiclass Classification. Explain the One-vs-One and One-vs-All strategies for multiclass classification. [5]

b) What is Support Vector Machine (SVM)? Explain the concept of maximum margin hyperplane and the role of kernel functions. [6]

c) Explain techniques for handling imbalanced data in classification problems. Discuss oversampling, undersampling, and SMOTE. [6]

Unit V — Unsupervised Learning — 18 Marks

Q5) Solve **any one** of the following:

a) Explain K-Means clustering with essential steps. Apply K-Means to cluster the following 2D points into K=2 clusters with initial centroids at (1,1) and (5,5): A(1,2), B(2,1), C(4,5), D(5,4), E(7,8). Show iterations until convergence. [6]

b) What is Outlier Analysis? Explain its importance, methods of detection (Isolation Forest, LOF), advantages and disadvantages. [6]

c) Write a short note on the Elbow method used in K-Means clustering. How does the Silhouette score complement it? [6]

OR

Q6) Solve **any one** of the following:

a) Explain Hierarchical clustering (agglomerative approach). Compare single-linkage, complete-linkage, and average-linkage with a dendrogram example. [6]

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b) Explain DBSCAN clustering. How does it handle noise and arbitrary-shaped clusters? Compare with K-Means. [6]

c) Write short notes on: [6]

- i) Spectral Clustering
- ii) Graph-Based Clustering

Unit VI — Introduction to Neural Networks — 17 Marks

Q7) Solve **any one** of the following:

a) Write a short note on the importance of Activation Functions in Neural Networks. Explain Sigmoid, Tanh, ReLU, and Softmax with their mathematical expressions and use cases. [5]

b) Explain Recurrent Neural Network (RNN) with an example. What is the vanishing gradient problem in RNNs and how does LSTM address it? [6]

c) Draw and explain the architecture of Convolutional Neural Network (CNN). Explain the role of convolutional, pooling, and fully connected layers. [6]

OR

Q8) Solve **any one** of the following:

a) Compare Backpropagation Network with Feedforward Network. Explain the backpropagation learning algorithm with the chain rule of derivatives. [5]

b) What are the different types of padding used in CNNs? Explain valid padding, same padding, and their effect on output dimensions. [6]

c) Write short notes on: [6]

- i) Convolutional Layer
- ii) Pooling Layer (Max, Average, Global)

*** End of Paper ***

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